25901743 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCNBC100 - BIOCHEMISTRY OF ENERGY TRANSFORMATION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which type of carrier protein moves one molecule in one direction? [K] / [CO1]

   a). Symport b). Antiport

   c). Uniport d). Bipolar

2. .....is a quantitative measure of the tendency of a redox pair to lose or gain electrons. [U] / [CO2]

   a). Redox Potential b). Oxidation state

   c). Activation energy d). Reduction factor

3. The lock and key model of enzyme action is also known as……. [U] / [CO3]

   a). Fischers theory

   b). Koshlands theory

c). Induced fit theory

d). substrate strain theory

4. The inactive protein part of the enzyme is called……. [K] / [CO3]

a). Apoenzyme b). holoenzyme

c). coenzyme d). isoenzyme

5. Which of the following high-energy compounds is an example for enol phosphate? [K] / [CO2]

a). Acetyl CoA b). ATP

c). Phosphoenolpyruvate d). Phosphocreatine

6. The primary mRNA transcript produced by RNA polymerase in eukaryotes is often referred to as ……. [K] / [CO3]

a). mRNA b). rRNA

c). hnRNA d). tRNA

[1x6 = 6]

Part B

Short Answer Questions

Answer any **2** questions

Each question carries **2** marks

7. Define Osmosis. [K] / [CO1]

8. Describe ATP-ADP cycle. [U] / [CO2]

9. Define oxidation. [K] / [CO2]

10. Define reaction specificity. [K] / [CO3]

[2x2 = 4]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

11. Describe transcytosis. [U] / [CO1]

12. Explain the various components of ETC. [K] / [CO2]

13. Explain the Watson Crick model of RNA. [K] / [CO3]

14. List down the factors affecting diffusion rate. [K] / [CO1]

15. Explain the stages of oxidation in food. [U] / [CO2]

16. Explain gene mapping and its classification. [K] / [CO3]

[5x4 = 20]

Part D

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain facilitated diffusion. [U] / [CO1]

18. Describe the methods of ATP synthesis and ATP-ADP cycle. [U] / [CO2]

19. Explain the major types of DNA damages and repair. [K] / [CO3]

20. Explain the process of protein biosynthesis in humans. [K] / [CO3]

[10x2 = 20]